

Week 1.8 — Debrief and Go Digital: Grading the First Splash, Then Opening the Electronics Lab

Sunday, August 9, 2026 · 3:00 – 5:00 PM · 14 students

Class 1.8 was the biggest turnout of the season — fourteen of us, 3:00 to 5:00, and no build. We started by grading our own work on the first splash, and we made ourselves write down both columns: six things that worked and six things that have to change. Then we looked at the entire season, start to finish, so everybody can see where this is going. We went through the company — why MATE runs the team like one, and what each position actually does — and every student wrote down their top three. Last hour we stood up an electronics lab: HackPacks and computer stations, and a first pass at the digital side of the house — Arduino Nano, servos, stepper motors, LED displays, and the IDE we'll write the code in. We closed on how the team is going to work together: GitHub for anything that's code, Google Drive for everything else.

14	6	6	1st
STUDENTS — BIGGEST TURNOUT YET	THINGS THAT WORKED	PROBLEMS NAMED FOR THE FIX LIST	DIGITAL ELECTRONICS LAB

FIRST-SPLASH RETROSPECTIVE

What worked

- Buoyancy devices did their job — the vehicle floated the way we intended.
- Center of mass below center of buoyancy. The ROV self-rights instead of rolling over.
- Tether waterproofing held. No intrusion through the tether run.
- Some of the motors worked — enough thrust to move the vehicle under control.
- The cart earned its place: hauled everything to the pool, then served as the topside control platform.
- The frame stayed together. Everything cut, drilled, and set-screwed in 1.5–1.7 held up in the water.

What needs addressed

- We can't tune buoyancy. It's whatever the foam gives us — no way to trim to neutral.
- The buoyancy foam won't stay attached. The mounting method has to change.
- One motor still only runs in one direction — the open Sabertooth fault from 1.7.
- The vehicle is slow, and we don't know yet whether that's thrust, drag, voltage, or all three.
- The control station takes deliberate effort — an interface and mixing problem, not a pilot problem.
- Motor placement may be wrong. Alternative thrust vectors are on the table.

WHAT WE LEARNED

- A retrospective only works if you write both columns. It's easy to list what broke; it matters just as much to write down what held, because that's what you don't want to accidentally redesign away.
- "It's slow" is not a diagnosis. Slow could be thrust, drag, voltage drop down the tether, or a motor not pulling its share. Naming the candidates is step one; measuring them is step two.
- "The control station required deliberate control" is an engineering finding, not a complaint about the pilot. When the operator has to fight the interface, the interface is what gets fixed.
- Buoyancy you can't adjust isn't really designed. A real vehicle has a way to trim to neutral and a mount that survives the water.
- Analog got us into the pool. Digital is what makes the vehicle controllable — a microcontroller between the joystick and the motors turns raw throttle into something a pilot can fly.

DECISIONS MADE

- Buoyancy becomes a tunable system, not a fixed one. We need a way to add and remove trim, and a mount that survives the water.
- Motor placement is reopened. Alternative thrust vectors get evaluated against real handling data instead of a sketch.
- The team goes digital — the Nano, servos, steppers, and displays are the path to a vehicle that can fly a mission.
- GitHub for code, Google Drive for everything else. Decided now, before there are forty files and no current copy.
- Every student named a top-three position. Elections come next; nobody gets a title without knowing the job.

ACTION ITEMS

Design a tunable buoyancy solution — adjustable trim plus a foam mount that stays attached in the water

Mechanical group · by 1.9

Trade-study alternative motor placements against the first-splash handling notes, not against a sketch

Mechanical group · by 1.9

Close out the one-direction motor fault with the replacement Sabertooth 2x5 and re-measure

Electrical group · by 1.9

Find the cause of the slow speed — separate thrust, drag, and voltage drop before changing anything

Electrical group · by 1.9

Collect every student's top-three position picks and set an election date

Coach John · by 1.9

Stand up the team GitHub org and the Drive folder structure; get every student an account and access

Coach John · by 1.9

Keep working the HackPack exercises between meetings — Arduino Nano, servo, stepper, and display

All students · ongoing

STUDENT LEARNING OUTCOMES

- Ran a structured engineering retrospective on their own vehicle — separating what held up from what has to change, and writing both down honestly.
- Saw the full season laid out end to end, so this stops being a series of Sundays and starts being a project with a deadline.
- Learned what each company position actually does, and picked a top three. Choosing a role you understand is different from volunteering for a title.
- First hands-on with digital electronics — Arduino Nano, servos, stepper motors, and LED displays on the HackPack stations, plus a first look at the IDE.
- First exposure to version control and shared storage as a team practice.

LOOKING AHEAD

Class 1.9 turns the fix list into hardware: the tunable-buoyancy design and foam mounting get worked, alternative motor placements get traded against what the first splash showed us, and the electrical group closes out the one-direction motor with the replacement Sabertooth. HackPack work continues in parallel — every student needs to be comfortable with the Nano before the code that flies this thing gets written. And elections are coming: everyone named a top three tonight.

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